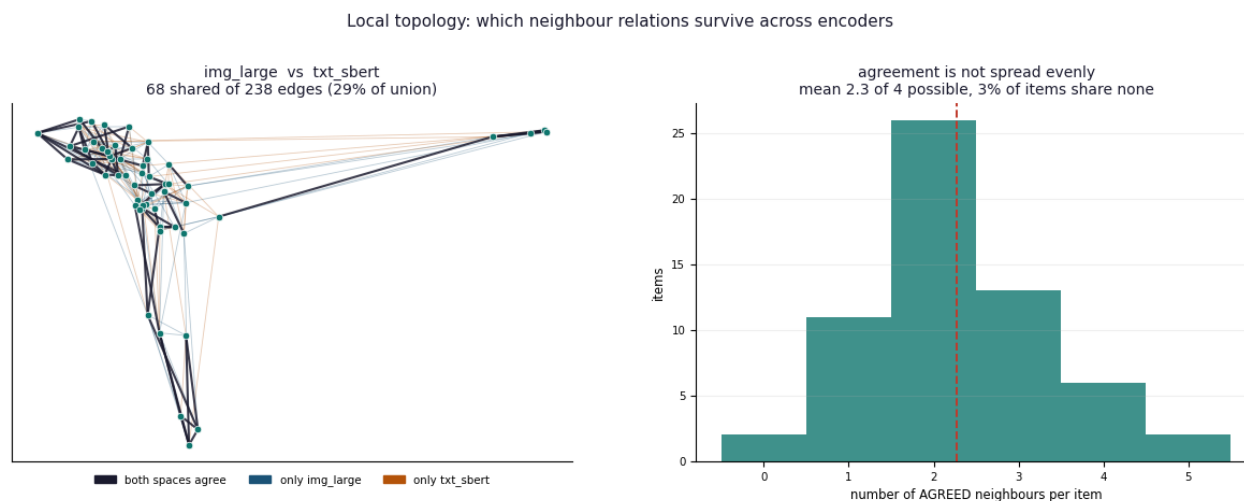


# Page 1 - Local topology: do neighbors survive across encoders?

The figure compares img\_large and txt\_sbert by asking whether each item keeps the same nearby items in both spaces.



## Test A - shared-neighbor graph

- Build a local k-nearest-neighbor graph in each encoder space, then overlay the two graphs on the same items.
- Dark edges mean both encoders agree that the same pair of items are neighbors; colored edges mean only one encoder creates that relation.
- Here, 68 of 238 union edges survive in both spaces: about 29% shared local topology between img\_large and txt\_sbert.

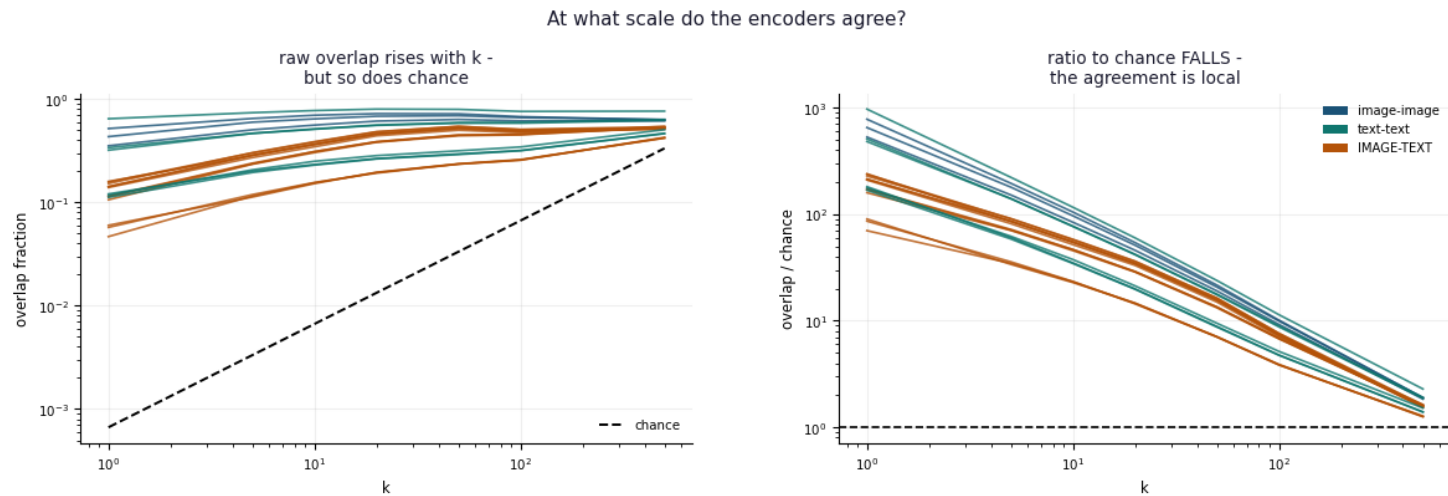
## Test B - per-item agreement histogram

- For each item, count how many of its possible neighbors are also neighbors in the other space.
- The mean is 2.3 agreed neighbors out of 4 possible, and only 3% of items share none.
- The key reading is heterogeneity: agreement is real, but it is not distributed evenly across all samples.

**Takeaway:** The two encoders do not share one identical global map, but they preserve a measurable part of local adjacency structure.

## Page 2 - Scale test: at what neighborhood size do encoders agree?

The second figure separates raw overlap from overlap normalized by chance as  $k$  grows.



### Test C - raw overlap vs $k$

- For  $k = 1, 5, 10, 20, 50, 100$ , and larger neighborhoods, compute the fraction of neighbors shared by two spaces.
- Raw overlap rises with  $k$  because larger neighborhoods include more items and therefore create more chances to overlap.
- Image-image pairs are strongest, text-text pairs are also strong, and image-text pairs are weaker but still above chance.

### Test D - overlap divided by chance

- Normalize the observed overlap by the expected random overlap at that same  $k$ .
- This ratio falls as  $k$  increases, which means the most meaningful agreement is concentrated in very local neighborhoods.
- The result supports local relational convergence more strongly than full global metric convergence.

**Interpretation** If raw overlap rises but the chance-normalized ratio falls, the encoders agree most strongly near each point. This is evidence for shared local topology, not necessarily for one universal coordinate system or one globally aligned metric.

# Page 3 - Sentence guide: what each test answers

Use these one-sentence descriptions in the report or presentation.

- **kNN overlap**

For each item, compare the k nearest neighbors in encoder A with the k nearest neighbors in encoder B; the score asks whether the two spaces preserve the same local topology.

- **Shared-edge graph**

Draw the neighbor edges from both encoders on the same set of items; edges that appear in both spaces are direct visual evidence of preserved adjacency relations.

- **Agreement histogram**

Count, item by item, how many neighbors are shared; this reveals whether alignment is broad across the dataset or concentrated in a subset of easier items.

- **Raw overlap across k**

Increase k and measure how the shared-neighbor fraction changes; this shows the scale at which the two encoders begin to agree.

- **Overlap/chance ratio**

Divide observed overlap by random-chance overlap at the same k; this prevents large k from looking strong merely because large neighborhoods overlap more easily.

- **CKA**

Compare centered Gram matrices rather than raw coordinates; CKA tests whether two spaces place the same samples near or far from one another at a global relational level.

- **Ridge map / linear stitching**

Fit one linear matrix from A to B on paired samples; good held-out retrieval or cosine means the target representation is linearly accessible from the source.

- **Orthogonal Procrustes**

Restrict the map to a rotation or reflection; success means the spaces are almost the same geometry in different coordinates, while failure implies scaling or anisotropy differences.

- **R@K retrieval**

Translate a source embedding and ask whether its true paired target appears among the top K nearest target vectors; this measures operational cross-space usability.

- **MLP comparison / control**

Compare the linear map with a nonlinear MLP and with chance or random baselines; if the MLP adds little, the relationship is essentially linear, and if controls fail, the effect is learned structure rather than artifact.

## Bottom-line claim

The figures support a careful version of the Platonic/representation-convergence idea: different encoders may share local relational structure even when their global metric geometry, dimensionality, anisotropy, and coordinate systems differ. The next decisive test is to combine kNN-overlap, CKA, Procrustes/Ridge mapping, and R@K into one compatibility matrix across all 7 spaces.